Watermark-Guided Tamper Localization and Recovery

A Key-Seeded Self-Embedding Fragile Watermarking Scheme with Joint Localization and Restoration Assessment

Tejas Mane[1] Shivam Singh[2] Amol Hipparkar[3] Guide: Dr. Deepti Varshney

Department of Computer Engineering, Ajeenkya DY Patil School of Engineering, Pune, India

[1]manetejas710@gmail.com [2]notshivamsingh@gmail.com [3]amolhipparkar04@gmail.com

1 The problem

A photograph is evidence only if you can show it was not edited after capture.

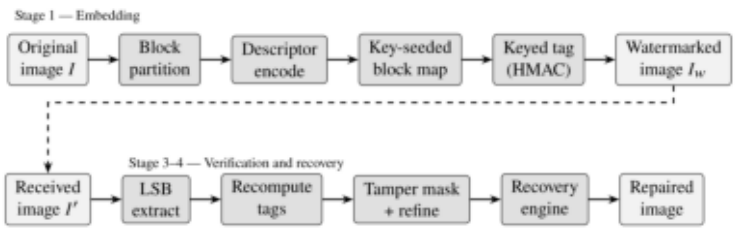
- A competent edit is invisible. A doctored image looks exactly as credible as a real one.
- Detectors trained on today's forgeries degrade on tomorrow's generator, and give no way to audit their verdict.
- Knowing that an image was altered is not enough. A court, an insurer or a newsroom needs to know where, and what was there before.

OBJECTIVE

Make the image carry its own proof and its own backup, so that any later edit is detected, localized to an 8x8 block, and repaired from data already inside the file itself.

No reference copy. No database. No training. No GPU. And nothing is invented: a block that cannot be honestly restored is marked black, never filled in with a guess.

2 How it works



1 PARTITION
into 8x8 blocks. The two low bit-planes of a block are 128 free bits of carrier.

2 DESCRIBE
each block in 96 bits: 12 zig-zag DCT coefficients quantized at JPEG quality-50 step sizes.

3 SCATTER
that descriptor to a partner block at least 0.25 min(M,N) away, picked by a key-seeded single-cycle permutation.

4 AUTHENTICATE
with a 32-bit truncated HMAC-SHA256 over the block's six upper bit-planes and the descriptor it carries.

5 VERIFY AND REPAIR
by recomputing every tag. A mismatch flags the block, which is rebuilt from the copy its partner holds.

3 The scheme

Carrier capacity, split per block

$$\mathrm{cap}(B) = 2B^2 = 128 \text{ bits} \;=\; \text{32-bit tag + 96-bit descriptor}$$

Keyed authentication tag

$$a_i = \mathrm{Trunc}_{32}[\mathrm{HMAC}_{K_s}(\mathrm{ID},\, i,\, \mathrm{MSB}(b_i),\, r_{m^{-1}(i)})]$$

$$\mathrm{MSB}(a) = a \;\wedge\; \texttt{0xFC}$$

Recoverability of a flagged set T

$$\rho = 1 - \frac{|U|}{|T|}, \qquad U = \{\, i \in T :\; m(i) \in T \,\}$$

$$\mathbb{E}[\rho] \;\approx\; 1 - \alpha \quad \text{(structural bound)}$$

One keyed hash and one 8x8 transform per block, so the whole pipeline is O(MN). No iteration, no optimization and no learned parameters: the scheme sits in complexity class P, and the same key reproduces the same bits every time.

4 Measured results

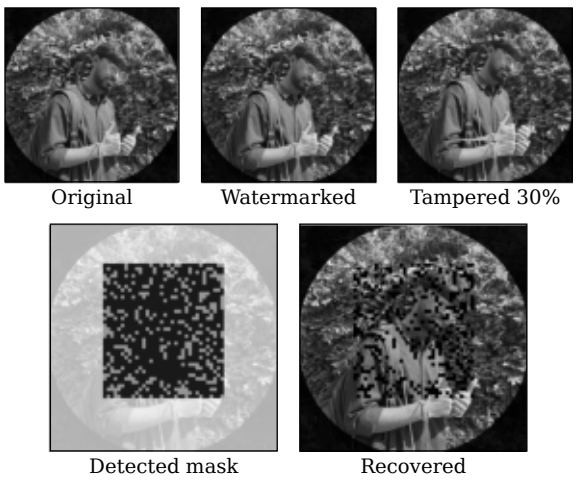
32 colour images (8 USC-SIPI + 24 Kodak), each pinned by SHA-256. 3,232 runs: 1,792 main grid (4 tamper classes x 7 ratios x 2 variants) + 320 null + 1,120 ablation. Automated sanity gate 25/27.

Watermarked PSNR / SSIM	**45.48 dB / 0.9885**
Block-level recall	**1.0000 (0.99998 inpainting)**
Pixel-level precision	**0.9599**
False positives in 1,802,240 checks	**0 (95% bound 1.7e-6)**
Mean recoverability rate	**0.6482**
In-region recovered PSNR	**28.49 dB (A) / 31.13 dB (C)**
Runtime, 512x512	**473 ms embed / 197 ms verify**

Reported honestly: block precision is 1.0000 only because the ground-truth rule counts a block as tampered when the region touches any of its pixels, so it is not evidence of detector quality. The zero false positives are a determinism check across images, variants and keys, not a measurement of the 2^-32 cryptographic bound.

5 On our own photograph

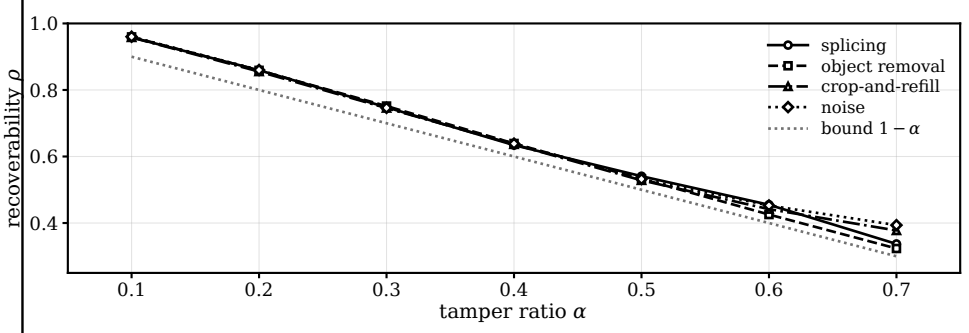
512x512 capture, copy-paste splicing over 30% of the frame, Variant A.



in the mask: black = flagged and repairable, grey = flagged but unrecoverable

Watermarked at 45.648 dB / SSIM 0.99356. Of 1,292 flagged blocks, 321 are unrecoverable (rho = 0.7515) and are left black rather than fabricated; pixel precision 0.9522, in-region PSNR 23.27 dB -- about 5 dB under the corpus mean, because dense foliage is what 12 DCT coefficients cannot hold. Shown as the worst case, not the best.

6 Recovery and conclusions



- Recoverability sits 3 to 6 points above its own structural bound at every ratio, falls monotonically, and never collapses.
- Recovered quality inside the tampered region is essentially flat across 10-70% coverage (29.81 dB mean), so mapping coverage, not descriptor fidelity, is the binding constraint.

Limits we accept: the scheme is fragile by design, so any re-encode destroys it; the 8x8 grid caps pixel precision near 0.96; and 4x4 blocks buy a finer mask for 2.85 dB and a weaker 16-bit tag.

Next: share each descriptor across several partners (fountain coding) to break the coverage ceiling.
github.com/BIGSUS24/watermark-recovery-project